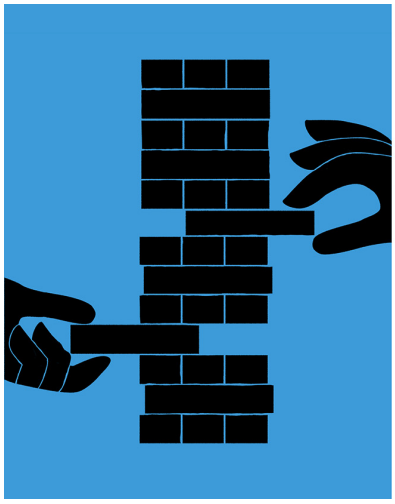


IMAGINE TRYING TO BUILD A HOUSE WHILE SOMEONE KEEPS STEALING YOUR BRICKS

OCEAN ACIDIFICATION

IN CONTEXT

KEY FIGURES

Kenneth Caldeira (1960–), **Michael E. Wickett** (1971–)

BEFORE

1909 Danish chemist Søren Sørensen develops the pH scale for measuring acidity.

1929 American biologists Alfred Redfield and Robert Goodkind discover that excess carbon dioxide in water suffocates squid.

1933 German chemist Hermann Wattenberg makes the first global survey of ocean acidity, as he analyzes results from the Atlantic expedition of the *Meteor* research vessel.

AFTER

2012 In the US, oceanographer James C. Zachos and his colleagues use fossil evidence from marine sediments to show that past acidification of the ocean has led to mass extinctions of sea creatures.

Adding carbon dioxide (CO_2) to the air not only triggers climate change but also makes the oceans more acidic. So far, the oceans have buffered the worst effects of global warming, absorbing up to half of the carbon dioxide